A Dense-Set Counterexample for Weak-Baire Measurability of the Norm

Independent research note

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Abstract

Question (B) of Avilés–Plebanek–Rodríguez asks whether relative weak-Baire measurability of a Banach-space norm on a norm-dense set forces weak-Baire measurability on the whole space. The answer is no in ZFC. For every nonseparable real Hilbert space X, we construct a norm-dense subset $A \subseteq X$ such that the restricted norm is the trace of a weak-Baire function depending on one linear coordinate, although the norm on X is not weak-Baire measurable. The parenthetical variant in which A must be a dense linear subspace is not resolved here.

Packet status

Run	`fa_banach_001`, lane 13.
Source	A. Avilés, G. Plebanek, and J. Rodríguez, *A weak* separable $C(K)^*$ space whose unit ball is not weak* separable*, arXiv:1112.5710, Section 3.3, Question (B), PDF page 22 [1].
Classification	Candidate counterexample, likely valid subject to human mathematical review.
Scope	Full negative answer to the norm-dense-set formulation; no claim for the dense-linear-subspace parenthetical.

The source question is reproduced here:

> (B) Let $(X, \|\cdot\|)$ be a Banach space and suppose $A \subseteq X$ is a norm dense set (or linear subspace) such that $\|\cdot\|$ is relatively $\mathrm{Ba}(X, w)$-measurable on A. Does this imply that $\|\cdot\|$ is $\mathrm{Ba}(X, w)$-measurable on X?
>
> Note that the analogous question for the property "B_{X^*} is w^*-separable" has positive answer (just apply the Hahn-Banach theorem), while for the property "X^* is w^*-separable" it has negative answer, see [5, Example 1.1].

In transcription: if $A \subseteq X$ is norm dense and the norm is relatively $\mathrm{Ba}(X, w)$-measurable on A, must the norm be $\mathrm{Ba}(X, w)$-measurable on X? The paper also asks the stronger-looking parenthetical variant where A is a linear subspace.

1 Proof intuition

Two elementary ideas fit together. First, every real function measurable for $\mathrm{Ba}(X, w) = \sigma(X^*)$ depends on only countably many dual coordinates. Countably many vectors cannot norm a nonseparable Hilbert space: a nonzero vector orthogonal to all of them is indistinguishable from zero. Hence the global Hilbert norm is not weak-Baire measurable.

Second, a dense subset need not retain any linear geometry. Split a Hilbert space as $\mathbb{R} \oplus_2 Z$. Choose a Borel function $\psi : \mathbb{R} \to [0, \infty)$ with dense graph and retain from each vertical fiber only the sphere $\|z\| = \psi(t)$. The resulting set is dense, while its norm is $\sqrt{t^2 + \psi(t)^2}$, a Borel function of the one weak coordinate t. Thus the restriction is measurable even though the global norm is not.

2 The counterexample

We first record the countable-dependence fact in the needed form.

Lemma 1 (Countable-coordinate dependence)**.** *Let $\mathcal{F}$ be any family of maps from a set Ω to $\mathbb{R}$. If $u : \Omega \to \mathbb{R}$ is measurable for $\sigma(\mathcal{F})$, then there is a countable subfamily $\mathcal{F}_0 \subseteq \mathcal{F}$ such that*

$$f(x) = f(y) \quad (f \in \mathcal{F}_0) \qquad \Longrightarrow \qquad u(x) = u(y).$$

Proof. Call a subset of Ω countably determined if it belongs to $\sigma(\mathcal{F}_0)$ for some countable $\mathcal{F}_0 \subseteq \mathcal{F}$. The countably determined sets form a sigma-algebra: for a countable union, take the union of the countably many subfamilies involved. This sigma-algebra contains every generator, so it is all of $\sigma(\mathcal{F})$.

For each $r \in \mathbb{Q}$, choose a countable subfamily determining $u^{-1}((-\infty, r))$, and take their union $\mathcal{F}_0$. Points having the same $\mathcal{F}_0$-coordinates belong to exactly the same rational sublevel sets of u, and therefore have the same u-value. □

Theorem 1. *Let X be any nonseparable real Hilbert space. There exists a norm-dense set $A \subseteq X$ such that $\|\cdot\||_A$ is measurable for the trace of* $\mathrm{Ba}(X, w)$ *on A, while $\|\cdot\| : X \to \mathbb{R}$ is not* $\mathrm{Ba}(X, w)$*-measurable.*

Proof. Choose a one-dimensional summand and write

$$X = \mathbb{R} \oplus_2 Z, \qquad \|(t, z)\| = \sqrt{t^2 + \|z\|^2},$$

where Z is nonseparable. Let $(q_n)_{n \geq 1}$ enumerate $\mathbb{Q} \cap [0, \infty)$, put

$$D_n = n\sqrt{2} + \mathbb{Q} \quad (n \geq 1), \qquad D_0 = \mathbb{R} \setminus \bigcup_{n \geq 1} D_n,$$

and define

$$\psi(t) = \begin{cases} q_n, & t \in D_n,\ n \geq 1, \\ 0, & t \in D_0. \end{cases}$$

The sets D_n are pairwise disjoint, countable, dense, and Borel; their complement D_0 is also Borel. Thus ψ is Borel. Its graph is dense in $\mathbb{R} \times [0, \infty)$: every open rectangle contains (t, q_n) after first choosing a rational q_n in its vertical interval and then $t \in D_n$ in its horizontal interval.

Define

$$A = \{(t, z) \in \mathbb{R} \oplus_2 Z : \|z\| = \psi(t)\}.$$

To see that A is norm dense, fix $(t, z) \in X$ and $\varepsilon > 0$. Choose q_n with $|q_n - \|z\|| < \varepsilon/2$, and choose $s \in D_n$ with $|s - t| < \varepsilon/2$. If $z \neq 0$, set $z' = q_n z/\|z\|$; if $z = 0$, choose $z' \in Z$ with $\|z'\| = q_n$. Then $(s, z') \in A$ and

$$\|(s, z') - (t, z)\| = \sqrt{|s - t|^2 + |q_n - \|z\||^2} < \varepsilon.$$

Let $\pi : X \to \mathbb{R}$ be the first-coordinate functional and put $\phi(t) = \sqrt{t^2 + \psi(t)^2}$. The function ϕ is Borel, hence $\phi \circ \pi$ is $\mathrm{Ba}(X, w)$-measurable. For every $(t, z) \in A$,

$$(\phi \circ \pi)(t, z) = \sqrt{t^2 + \psi(t)^2} = \sqrt{t^2 + \|z\|^2} = \|(t, z)\|.$$

Therefore the restricted norm on A is measurable for the trace of $\mathrm{Ba}(X, w)$.

It remains to show that the global norm is not weak-Baire measurable. For a Banach space, $\mathrm{Ba}(X, w) = \sigma(X^*)$. If the norm were measurable, Lemma 1 would give $x_n^* \in X^*$, $n \geq 1$, on whose joint coordinates it depends. By the Riesz representation theorem, the x_n^*'s are represented by a countable family of Hilbert vectors. Their closed linear span is separable and hence proper in nonseparable X. Choose $0 \neq v$ orthogonal to it. Then $x_n^*(v) = x_n^*(0)$ for every n, so countable dependence gives $\|v\| = \|0\| = 0$, a contradiction. □

3 Scope, verification, and novelty

The construction is ZFC and already works in $X = \mathbb{R} \oplus_2 \ell_2(\omega_1)$. It answers the source implication negatively when A is an arbitrary norm-dense set. It deliberately does not make A linear. Requiring linearity turns the construction into the dense graph of a discontinuous operator over a countably coordinatized quotient; no valid construction or impossibility proof for that variant was found in the upgrade audit.

The proof was checked at four pressure points: the cosets $n\sqrt{2} + \mathbb{Q}$ are pairwise disjoint; the graph of ψ is dense; radial rescaling gives the displayed exact distance formula; and Lemma 1 applies to the rational sublevel sets of the norm. No computational assumption or set-theoretic hypothesis beyond ZFC is used.

A bounded search on August 13, 2026 covered the run indexes, arXiv:1112.5710, the exact text of Question (B), the phrases *relatively* $\mathrm{Ba}(X, w)$*-measurable*, *norm dense set*, and *weak Baire norm*, plus the title and authors. It found the source and nearby work on Baire structures in $C(K)$, but no later explicit answer and no matching shell construction. Novelty confidence is moderate because the proof is elementary and may be folklore. Human review should focus on literature novelty and on the intended parsing of the source's parenthetical *(or linear subspace)*; the mathematical counterexample itself is self-contained.

References

[1] A. Avilés, G. Plebanek, and J. Rodríguez, *A weak* separable $C(K)^*$ space whose unit ball is not weak* separable*, arXiv:1112.5710 (2011).